

Mann Lohchab

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SUMMARY

Systems and backend engineer with hands-on experience building production APIs, distributed systems, and low-level infrastructure using C, Rust, and Zig. Experienced in payments-adjacent ledger systems, asynchronous services, databases, networking, and Linux-based infrastructure. Built a trading ledger in Zig with N-API integration to Node.js, along with Redis-like, object storage, message queue, and Raft implementations in Rust.

EXPERIENCE

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|---|----------------|
| BhaaratMind
<i>Foundational Backend Engineer</i> | 2026 – Present |
| <ul style="list-style-type: none">– Lead backend architecture for AI products supporting workflows across 22+ languages.– Build scalable APIs, data services, and multilingual AI workflows for products including Manthan AI.– Own backend reliability, performance, and production systems.– Stack: TypeScript, Python, FastAPI, PostgreSQL, Redis, Docker. | |
| GhostWay
<i>Co-founder</i> | 2026 – Present |
| <ul style="list-style-type: none">– Built technical and product foundations for food discovery and delivery workflows connecting kitchens, customers, and riders.– Worked across product architecture, operations, backend systems, and service design for the Sonipat/Kundli market. | |
| KMOS
<i>Co-founder & Adviser</i> | 2024 – Present |
| <ul style="list-style-type: none">– Led architecture and engineering delivery for web, application, and systems-focused client projects.– Established engineering standards and provided long-term technical ownership beyond initial product launches. | |

TECHNICAL SKILLS

Languages: C, Rust, Zig, TypeScript, JavaScript, Python, Bash, Java

Backend: Node.js, Express.js, FastAPI, Bun, Tokio, REST APIs, N-API, Asynchronous Programming

Frontend: Svelte, React.js, React Native, TailwindCSS, HTML, CSS

Databases & Storage: PostgreSQL, MongoDB, SQL, Redis, Object Storage

Systems: Linux, BSD, POSIX, Unix Systems Programming, Process Management, IPC, TCP/IP

Distributed Systems: Raft, Message Queues, Distributed Storage, Concurrency, Consensus Systems

Infrastructure: Docker, Podman, Kubernetes, Firecracker, LXD, QEMU, Dokploy, Nginx

Networking: MQTT, TCP/IP, HTTP, REST, Network Services

SYSTEMS & BACKEND PROJECTS

- | | |
|--|------------------|
| Trading Ledger / zledger | Zig, Rust, N-API |
| <ul style="list-style-type: none">– Built a trading ledger and orderbook engine in Zig supporting order validation, balance management, order matching, and ledger operations.– Implemented a native N-API integration exposing the Zig engine to Node.js applications.– Integrated the native engine with Rust and implemented a database writer for persisting ledger-related state.– Worked with concurrency, native memory, FFI boundaries, and systems-level state management. | |
| Redis-like In-Memory Server | Rust, Tokio |

- Implemented a Redis-inspired in-memory server in Rust using Tokio for asynchronous networking and concurrent task execution.
- Worked with TCP connections, asynchronous I/O, shared state, concurrent clients, and event-driven server architecture.

S3-style Object Storage

Rust

- Built an S3-inspired object storage service in Rust to explore object storage APIs, request handling, and persistent data management.
- Designed storage abstractions and HTTP-based object operations.

Message Queue

Rust

- Implemented a message queue in Rust with producer and consumer workflows and asynchronous message processing.
- Explored concurrent message handling, delivery semantics, and distributed messaging architecture.

Raft Consensus Implementation

Rust

- Implemented the core Raft consensus algorithm to study distributed state replication and fault tolerance.
- Worked with leader election, terms, replicated logs, node coordination, and consensus state.

POSIX Unix Shell

C, POSIX

- Implemented a Unix shell in C using POSIX system interfaces for command execution and process management.
- Worked directly with `fork()`, `exec()`, pipes, file descriptors, and Unix processes.

Enterprise Resource Planning (ERP) System

TypeScript, Node.js, PostgreSQL

- Engineered a full-stack ERP platform for managing student, financial, administrative, and institutional workflows.
- Designed RESTful APIs, role-based authentication and authorization, and modular backend services.
- Designed PostgreSQL schemas and optimized relational queries for reliable data management.

SYSTEMS & INFRASTRUCTURE

Operating Systems: Strong practical experience with Linux, BSD, POSIX APIs, Unix tooling, Bash, and system-level debugging.

Virtualization: QEMU, Firecracker, and LXD; experience exploring lightweight virtual machines and isolated workloads.

Containers & Deployment: Docker, Podman, Kubernetes, Dokploy, and Nginx.

Networking: TCP/IP, HTTP, MQTT, REST services, and asynchronous network programming.

Embedded / Network Systems: RutOS and MQTT-based systems.

EDUCATION

SRM University

Expected 2028

B.Tech in Computer Science & Engineering (Core)

CGPA: 7.5

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming.

Central Board of Secondary Education (CBSE)

Class XII – Science (Physics, Chemistry, Mathematics, Biology)

87%

Central Board of Secondary Education (CBSE)

Class X

89.16%